Classifier evaluation

INF-GRAM — Machine Learning for Healthcare

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September 16, 2026

Session roadmap

Why evaluation deserves a full lecture

In healthcare, evaluating a classifier is not only a technical step. It is directly linked to the clinical question:

- which errors matter most?
- how rare is the target condition?
- do we need ranking, probabilities or hard decisions?

Learning objectives

By the end of the lecture, you should be able to:

- read a confusion matrix;
- interpret sensitivity, specificity, precision and recall;
- explain the meaning of F1-score;
- compare ROC and precision–recall curves;
- describe a correct train/validation/test workflow.

Start from the clinical problem

The metric depends on the use case

Before choosing a metric, ask:

- Is the positive class rare?
- Is a false negative more serious than a false positive?
- Do we care about ranking or about a fixed threshold?
- Will the model be used for triage, diagnosis or screening?

There is no universally “best” metric.

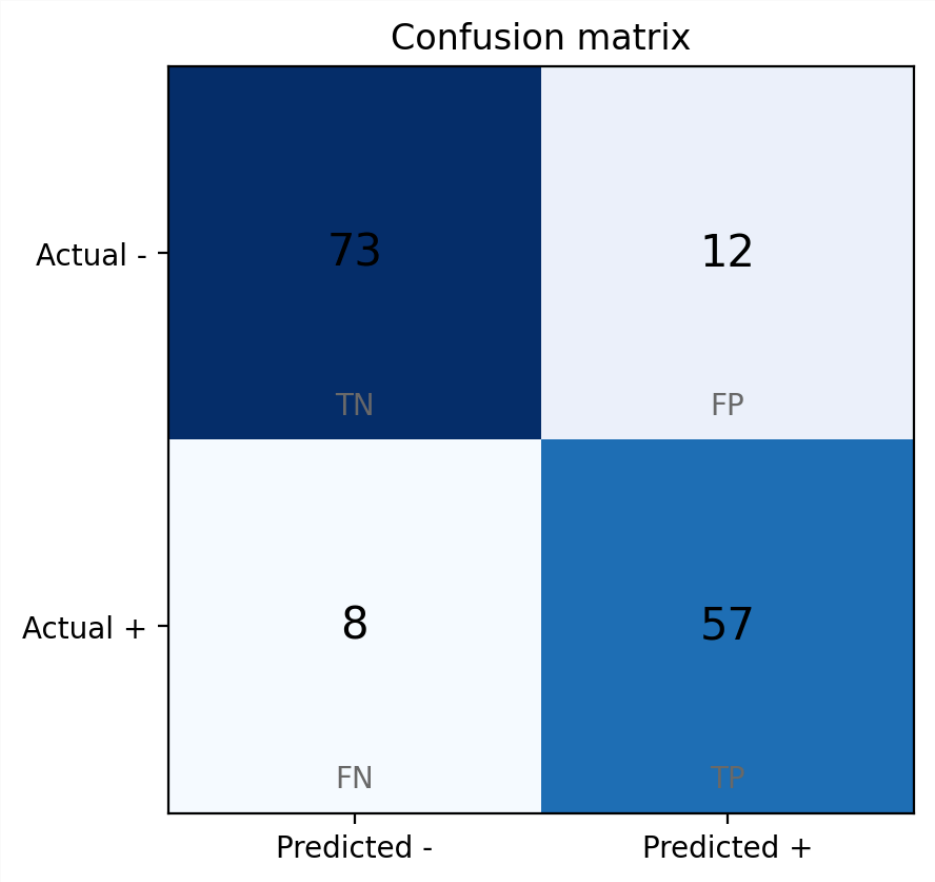
Confusion matrix and threshold-based metrics

The confusion matrix

For binary classification, every prediction belongs to one of four categories:

- true positive (TP)
- false positive (FP)
- true negative (TN)
- false negative (FN)

These four counts define many of our evaluation metrics.



Four key metrics

- **Sensitivity / recall**: (TP / (TP + FN))
- **Specificity**: (TN / (TN + FP))
- **Precision**: (TP / (TP + FP))
- **Accuracy**: ((TP + TN) / (TP + TN + FP + FN))

What these metrics mean in words

- **Sensitivity**: among true positives, how many did we detect?
- **Specificity**: among true negatives, how many did we correctly reject?
- **Precision**: among predicted positives, how many are truly positive?
- **Accuracy**: what proportion of decisions are correct overall?

Why accuracy can be misleading

If the positive class is rare, a classifier can achieve high accuracy simply by predicting the majority class most of the time.

So in imbalanced settings, accuracy is often a poor summary.

The F1-score

The F1-score combines precision and recall:

[F_1 = 2 { + }.]

It is useful when both missed positives and false alarms matter.

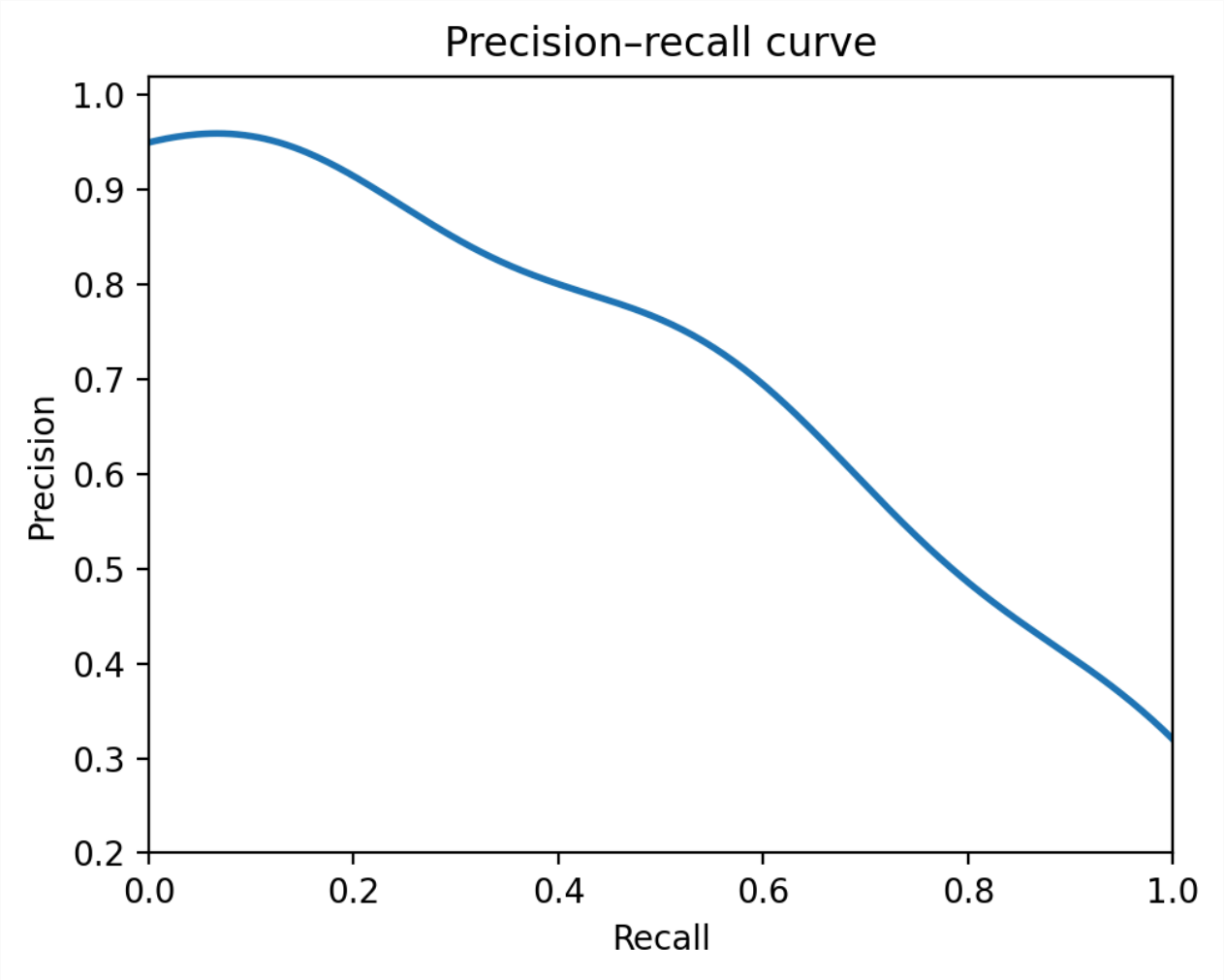
Threshold-free views of performance

Precision–recall curve

A precision–recall curve is especially informative when:

- the positive class is rare;
- retrieving positives matters a lot;
- false positives are costly.

Example of a PR curve



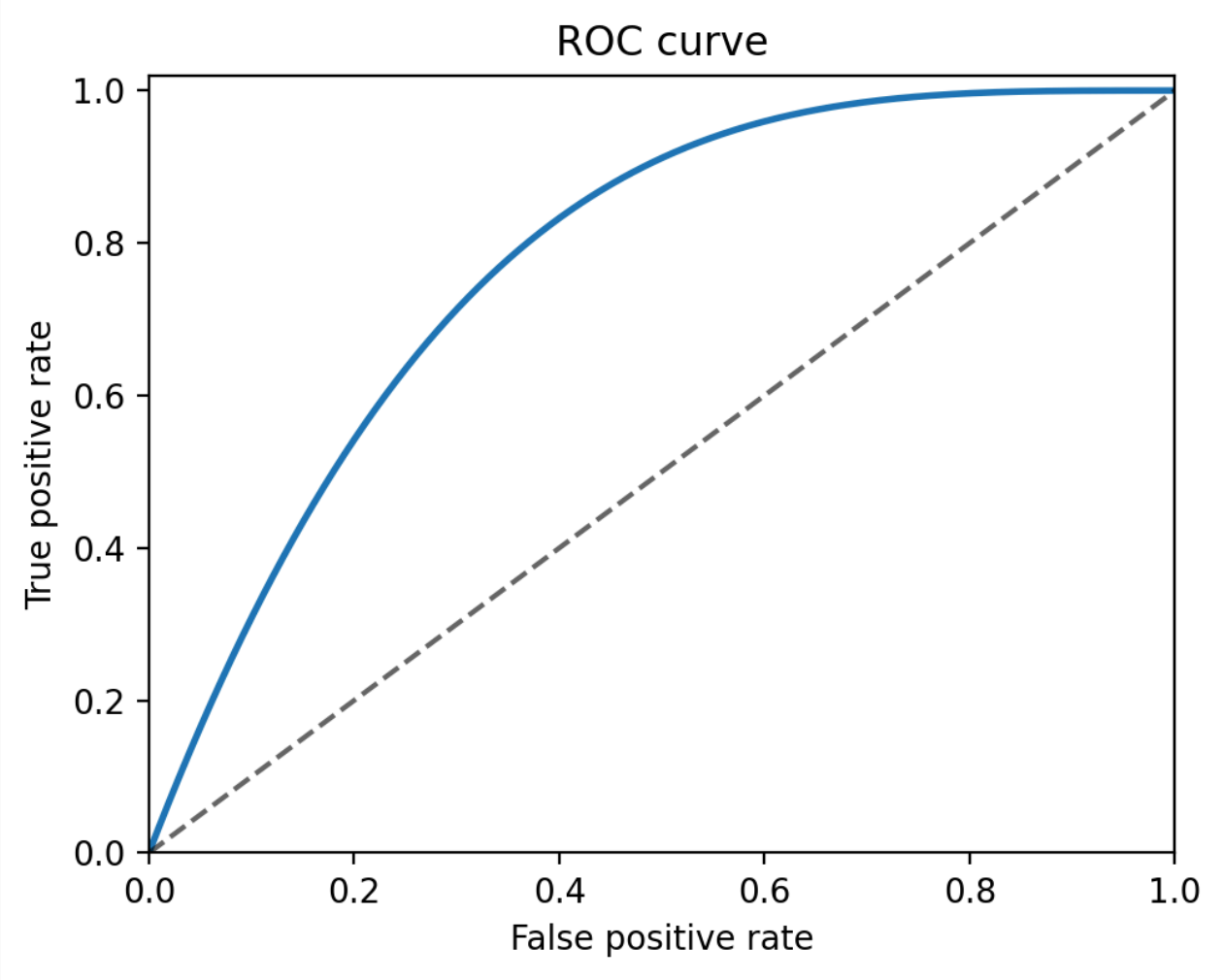
ROC curve

The ROC curve plots:

- true positive rate (sensitivity) on the y-axis;
- false positive rate on the x-axis.

The area under the ROC curve (ROC AUC) summarizes ranking quality over all thresholds.

Example of a ROC curve



ROC or precision–recall?

- **ROC** is general and widely used.
- **PR** is often more informative for rare outcomes.
- Neither curve replaces thinking about the actual decision threshold.

A metric that matters in healthcare: calibration

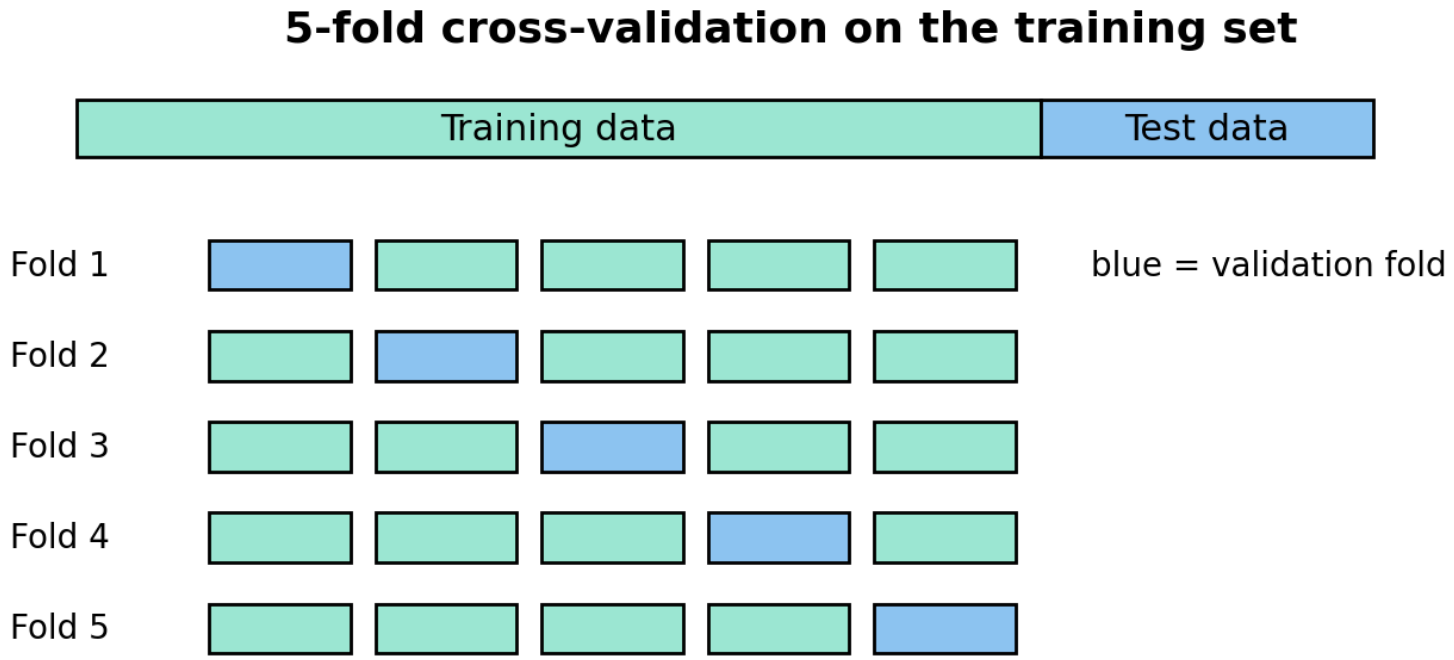
Good discrimination is not enough. We often also care whether predicted probabilities are **well calibrated**.

Example:

- among patients predicted at 20% risk,
- do about 20% truly experience the outcome?

Validation workflow

A correct train / validation / test strategy



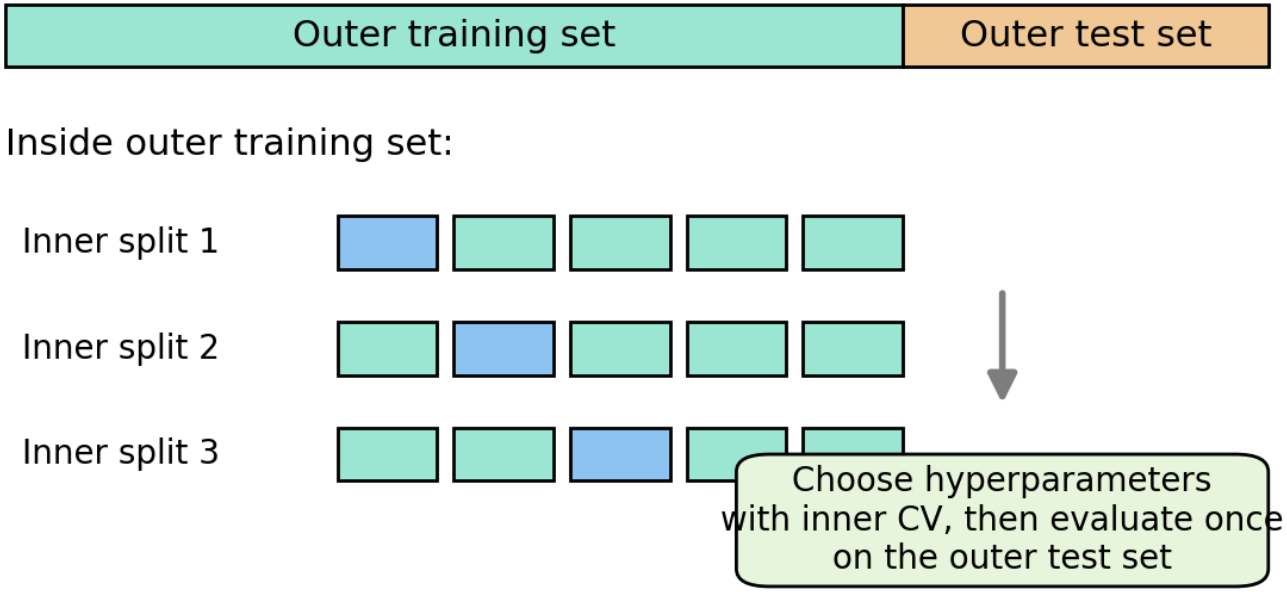
The golden rule

The test set must be used **once**, at the end, for final evaluation.

If we repeatedly inspect the test performance while choosing models, then the test set stops being a true test set.

Nested validation for model selection

Nested validation separates model tuning from final evaluation



Typical evaluation mistakes

Avoid the following:

- tuning hyperparameters on the test set;
- reporting only the best metric and hiding the rest;
- ignoring class imbalance;
- forgetting calibration;
- comparing models without a fair validation scheme.

Take-home messages

What to remember from this lecture

- Evaluation metrics must match the clinical use case.
- Sensitivity, specificity, precision and recall answer different questions.
- ROC and PR curves summarize performance across thresholds.
- A correct validation workflow is essential to avoid optimistic results.
- In healthcare, calibration and clinical interpretability also matter.